

THE STORY OF NAVIGATION

THE QUEST TO EXPLORE AND MAP THE WORLD'S OCEANS AND CONTINENTS

Perhaps the most surprising fact in the history of navigation is that, until the 18th century, it was impossible for explorers and mariners to determine their position accurately. Today, thanks to developments in navigational technology, it is possible to pinpoint locations to within a few meters.

The earliest sailors had no means of accurate navigation other than by sight, relying on landmarks along coastlines, judging distances and directions from the positions of the Sun, Moon, and stars, and using simple sounding devices, such

as weighted lines, to keep ships from running aground. The invention of instruments such as the magnetic compass, astrolabe, and sextant

John Harrison

English clockmaker John Harrison was the first to make accurate timepieces that enabled longitude to be calculated with precision.

enabled direction and latitude to be gauged reasonably accurately (by measuring the angle of the Sun or a star above the horizon) but the problem remained of how to determine longitude.

ACCURATE NAVIGATION

Calculating longitude depends on comparing local time with “universal” time (the time at an agreed location, which is now Greenwich, England). Each hour’s difference equates to 15 degrees’ difference in longitude. Calculating longitude therefore relies on accurate timepieces, which did not exist until John Harrison developed his chronometer in the 18th century. The next major advances in navigation did not come until the 20th century, with the advent of the gyroscopic compass, radar, and, from the 1990s, of the global positioning system (GPS).



H1 chronometer

LATITUDE AND LONGITUDE

LATITUDE

Latitude lines (parallels) run horizontally on a map and are measured in degrees north or south of the equator. Each degree is about 69 miles (111 km) apart.



LONGITUDE

Longitude lines (meridians) run vertically on a map and are measured in degrees east or west of Greenwich, England. They meet at the poles and are farthest apart at the equator.



“ONE OF THE MOST EXQUISITE MOVEMENTS EVER MADE.”

William Hogarth, English artist, on Harrison's H1 chronometer, from *Analysis of Beauty*, 1753

3000–1500 BCE

Early sounding

Ancient Egyptians use sounding reeds to measure water depth and gauge their position from coastal landmarks.



12th-dynasty sailing boat

11th century

Dead reckoning

Sand clocks are used for dead reckoning: measuring the time traveled and speed to estimate a vessel's position.



Sand clock

1300–1500

Navigational charts

Portolan charts of the Mediterranean and European coastlines allow sailors to navigate from port to port using compass bearings.



Portolan chart

c. 150

Ptolemy's maps

A Roman based in Egypt, Ptolemy creates maps using a grid system that influenced navigational maps until the 17th century.



Ptolemy's map

c. 1100

The compass

Chinese sailors are the first to use a magnetic compass (which uses a magnetized needle to show the direction of north and south) for navigation.



Mariner's compass

c. 1480

The astrolabe

Sailors start to use astrolabes to estimate latitude by measuring the angle of the Sun or a particular star above the horizon.



Mariner's astrolabe